AI-Based Voice-Controlled Smart Home
Functional Use Case Diagram Design Proposal

Initial Demonstration Scope

Voice command
recognition
validation
device resolution
device execution
feedback

Initial target:
lamp or smart plug with
explicit ON/OFF commands.

Training, dataset preparation
and hardware setup are treated
as development activities.

Invalid or uncertain commands terminate the control scenario before device actuation.

Uses mappings configured through UC04.

Resolution does not modify the configuration.

Device state may be:

- Requested
- Confirmed
- Unknown

Voice-Controlled Smart Home System

Speech Recognition

UC12
Recognize Voice Command

«include»

«include»

Home Automation

«include»

UC02
View Device Status

UC01
Control Device by Voice

UC03
Configure Supported Devices

UC04
Define Command-to-Device Mappings

End User

System Configurator

Control and Integration

UC10
Reject Invalid or Uncertain Command

«extend»
[invalid or uncertain command]

UC05
Validate Recognized Command

«include»

UC06
Resolve Target Device and Action

«extend»
[status-read failure]

UC08
Read Device State

«include»

UC09
Provide Operation Feedback

«include»

UC11
Handle Device or Communication Failure

«include»

UC07
Execute Device Command

«include»

«extend»
[execution failure]

Possible outcomes:

- Confirmed success
- Rejected command
- Detected failure
- Unconfirmed execution

Recognizes a supported command from captured audio.

Inference location remains open.

Smart Device

Work Package	Lead
WP1 - ML and Speech Recognition	Saad
WP2 - Home Automation Software	Zakaria
WP3 - Hardware and Electronics	Saad
WP4 - Embedded Software and Integration	Zakaria
Interfaces and End-to-End Evaluation	Joint